



HWA CHONG INSTITUTION
Preliminary Examination
Higher 1

CANDIDATE
NAME

CT GROUP

12S

CHEMISTRY

8872/02

Paper 2

18 September 2013

2 hours

Candidates answer **Section A** on the Question Paper.

Additional Materials: Data Booklet

Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue, correction fluid or tapes.

Section A

Answer **all** questions.

Section B

Answer **two** questions on separate answer paper.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINERS' USE ONLY

Paper 1	Paper 2		TOTAL
Multiple Choice	Section A (Structured)	Section B (Free Response)	110
	Q1 / 8	Q5 / 20	
	Q2 / 15	Q6 / 20	
	Q3 / 5	Q7 / 20	
	Q4 / 12		
/ 30	Subtotal / 40	Subtotal / 40	

Section A

Answer **all** the questions in this section in the spaces provided.

- 1 (a)** The elements of the third period of the Periodic Table, sodium to sulfur, all form chlorides by direct combination.

- (i) Sulfur forms a number of chlorides which are liquid at room temperature. Which other element of the third period forms a chloride which is liquid at room temperature?

.....

Aluminium chloride may be produced by passing a stream of chlorine over heated aluminium powder in a long hard-glass tube.

- (ii) Write a balanced equation, with state symbols, for this reaction of aluminium with chlorine.

.....

- (iii) No chloride of argon has ever been produced. Suggest a reason for this.

.....

.....

[3]

- (b)** When chlorides of the elements of the third period are added to water, some simply dissolve while others can be seen to react with the water.

Write equations for the reactions of the chlorides with water.

Sodium chloride

.....

Aluminium chloride

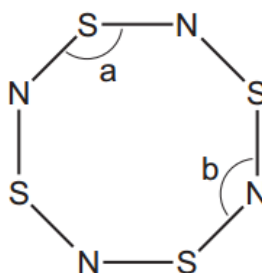
.....

Silicon tetrachloride

.....

[3]

- (c) Sulfur forms the compound S_4N_4 with nitrogen. The structure of S_4N_4 is shown below. Assume all bonds shown are single bonds.



- (i) Determine the number of lone pairs of electrons around a nitrogen atom and a sulfur atom in S_4N_4 .

Nitrogen atom:

Sulfur atom:

- (ii) Which bond angle, a or b, in the S_4N_4 molecule will be smaller? Explain your answer.

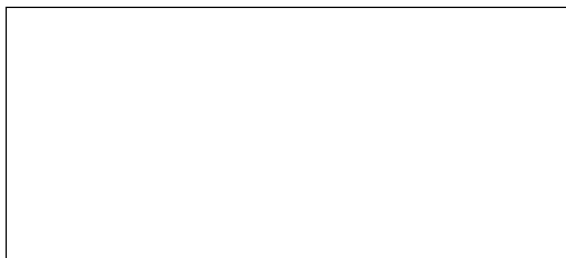
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.....

[2]
[Total: 8]

- 2 (a) E300 is an antioxidant used in white wines. It prevents dissolved oxygen reacting with the ethanol to form an acid, **X**, which would produce a sour-tasting wine.

In the box below draw the displayed formula of the acid, **X**, responsible for the wine's sour taste.



[1]

- (b) A student attempted to oxidise ethanol to acid **X** in the laboratory.

Give the chemical reagents and reaction conditions the student would need to use.

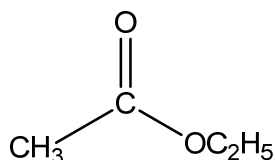
Reagents:

Conditions:

[2]

- (c) Use the table of characteristic values for infra-red absorption in the *Data Booklet* to answer this question.

Infra-red absorptions can be used to identify functional groups in organic compounds. For example, ethyl ethanoate shows absorptions at $1000\text{--}1300\text{ cm}^{-1}$ and $1680\text{--}1750\text{ cm}^{-1}$.



ethyl ethanoate

The infra-red spectrum of the purified product the student actually obtained has strong absorptions at about $1720\text{--}1740\text{ cm}^{-1}$; it has no broad peak at $2500\text{--}3200\text{ cm}^{-1}$.

Use the *Data Booklet* together with the information given to identify the product made by the student. Give your reasoning.

Name or formula of product:

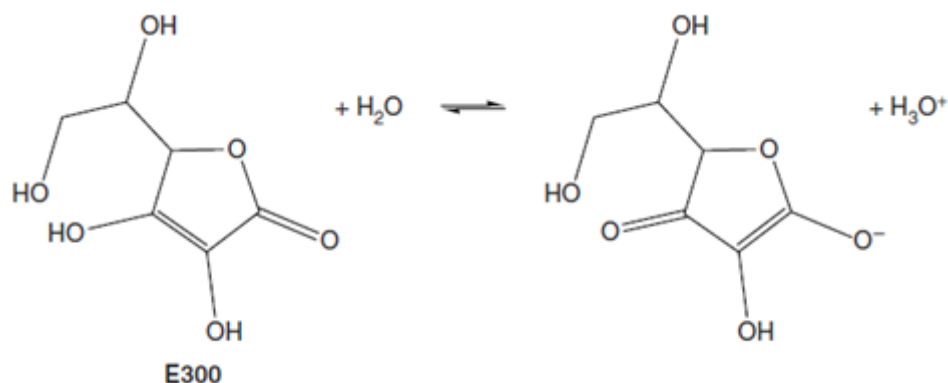
Reasoning:

.....

.....

[3]

- (d) When E300 is added to water, an acid–base equilibrium is set up, as shown below.



- (i) Explain what is meant by the term *base* in the Bronsted–Lowry theory.

.....

- (ii) Circle **two** species in the equation above that are acting as bases.

- (iii) When a solution of E300 reacts with calcium carbonate, fizzing occurs and a solid product can be extracted from the resulting mixture.

Identify the gas produced and suggest a formula for the solid product.

Gas:

Formula of solid product:

[4]

- (e) E300 has a C=C bond with two different groups on each carbon. It does not, however, show geometrical isomerism whereas 1,2–dichloroethene does.

Explain why 1,2–dichloroethene shows geometrical isomerism **and** suggest a reason why E300 does not.

.....

.....

.....

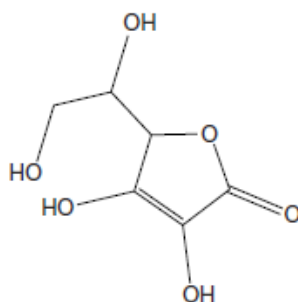
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.....

[2]

(f) A primary alcohol group in E300 reacts with $C_{17}H_{35}COOH$ to form another antioxidant.

(i) Circle a primary alcohol group on the structure of E300 below.



E300

(ii) Draw the structural formula of the new functional group formed in this reaction.

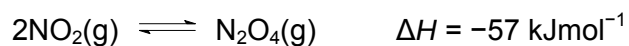
(iii) What else must be added to a mixture of E300 and $C_{17}H_{35}COOH$, to make the new antioxidant?

.....

[3]

[Total: 15]

- 3 (a) Dinitrogen tetroxide, $\text{N}_2\text{O}_4(\text{g})$, and nitrogen dioxide, $\text{NO}_2(\text{g})$, coexist in the following *dynamic equilibrium*.



A chemist adds 4.00 moles of NO_2 to a container with a volume of 2.00 dm^3 . The container is sealed, heated to a constant temperature and allowed to reach equilibrium.

The equilibrium mixture contains 3.20 moles of NO_2 .

- (i) Explain what is meant by the term *dynamic equilibrium*.

.....

- (ii) Write an expression for the equilibrium constant K_c for this reaction.

- (iii) Calculate the value of K_c under these conditions, giving the units.

[4]

- (b) The experiment is repeated but the temperature is increased.

State the effect on the concentrations of NO_2 and N_2O_4 when the mixture reached equilibrium.

.....

[1]

[Total: 5]

- 4 (a) Simple esters are flammable liquids. Flammability is affected by volatility.

Write the following homologous series in order of boiling point, assuming molecular masses are similar.

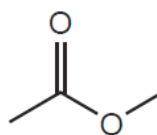
alcohols alkanes esters

highest boiling point

 lowest boiling point

[1]

- (b) (i) The structure of methyl ethanoate, $C_3H_6O_2$, is shown below.



Write an equation for the complete combustion of methyl ethanoate.

.....

- (ii) Define *standard enthalpy change of formation*.

.....

- (iii) Use the standard enthalpy changes of combustion, $\Delta H_c^\ominus$, in Table 1 to calculate the standard enthalpy change of formation of methyl ethanoate.

Table 1

	$\Delta H_c^\ominus$, / kJ mol ⁻¹
carbon	-393.5
hydrogen	-285.8
methyl ethanoate	-1592.1

..... kJ mol⁻¹

[5]

- (c) A student used the apparatus shown in Figure 1 to carry out experiments to determine the standard enthalpy change of combustion for methyl ethanoate.

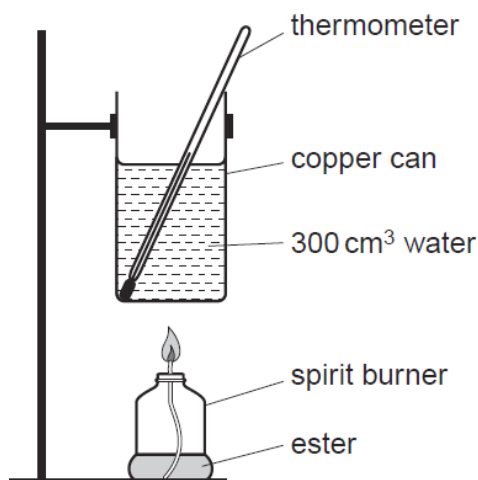


Figure 1

An initial experiment was carried out using methyl ethanoate. This ester was combusted in a spirit burner underneath a copper can, which has a mass of 250 g, so that the flame from the burner heated 300 cm³ of water in the can. It was found that 0.980 g of ester was required to raise the temperature of the water in the can by 10.0 °C.

Calculate the total thermal energy in kJ gained by the water and the copper can in this initial experiment.

The specific heat capacities of water and copper are 4.18 and 0.384 J g⁻¹ K⁻¹, respectively.

Take the density of water to be 1.00 g cm⁻³. Assume that the water and copper are in thermal equilibrium with each other. Express your answer to the appropriate number of significant figures.

Thermal energy added to water =

Thermal energy added to copper =

Total thermal energy =

..... kJ
[3]

- (d) Using the ΔH_c° value in Table 1, calculate the total theoretical thermal energy in kJ released by the mass of methyl ethanoate combusted in this initial experiment.

..... kJ
[2]

- (e) In terms of the ease of lighting, how does methyl ethanoate compare to decyl ethanoate ($\text{CH}_3\text{COOC}_{10}\text{H}_{21}$)?

.....
.....

[1]

[Total: 12]

11
Section B

Answer **two** questions from this section on separate answer paper.

- 5 (a) Chlorine consists of two isotopes, ^{35}Cl and ^{37}Cl , in the abundance ratio 3:1.
- (i) Write the full electronic configuration of chlorine.
 - (ii) Define the term 'isotope'.
 - (iii) State the number of protons, neutrons and electrons in a ^{37}Cl atom.
 - (iv) Calculate the relative atomic mass of chlorine.
 - (v) Sketch the trend of the successive ionisation energies of the first 10 electrons of a chlorine atom. Briefly explain the shape of your sketch.
- [7]
- (b) Chlorine forms compounds with almost all of the elements, exhibiting oxidation numbers from -1 to $+7$.
- (i) Explain in terms of oxidation numbers the reaction(s) undergone by chlorine in the following equation.
$$\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$$
 - (ii) Describe, in terms of orbital overlap, the bonding in a HCl molecule. Draw a clearly labelled diagram to illustrate your answer.
- [4]
- (c) A solution of HCl has a pH of 1.3 while a solution of $\text{CH}_3\text{CO}_2\text{H}$, at the same concentration, has a pH of 3.0.
- (i) Calculate the concentration of H^+ ions in both the acid solutions. Explain any difference in your answers.
- Both hydrochloric acid and $\text{CH}_3\text{CO}_2\text{H}$ react with magnesium metal, giving off hydrogen gas. For a fixed amount of magnesium, the rate equation for the reaction is as follows.
- $$\text{rate} = k [\text{H}^+(\text{aq})]$$
- When 20.0 cm^3 of each of the acid solutions in part (c)(i) is separately reacted with an excess of magnesium, the same volume of hydrogen is given off, but $\text{CH}_3\text{CO}_2\text{H}$ solution reacts more slowly.
- (ii) Write an equation for the reaction between $\text{CH}_3\text{CO}_2\text{H}$ and Mg .
 - (iii) Calculate the volume of hydrogen given off at room temperature and pressure.
 - (iv) Explain why the hydrogen gas is evolved more slowly from the $\text{CH}_3\text{CO}_2\text{H}$ solution but why, eventually, it produces the same volume of hydrogen gas.
- [6]
- (d) A solution containing $\text{CH}_3\text{CO}_2\text{H}$ and its sodium salt acts as an acidic buffer. Buffers are also present in human blood.
- Explain what is meant by an *acidic buffer*. Use blood as an example to illustrate your answer.

[3]

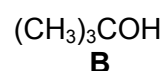
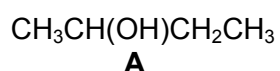
[Total: 20]

6 (a) The oxides of the Period 3 elements include the following: Na_2O , Al_2O_3 and SO_2 .

- Draw dot-and-cross diagrams to illustrate the bonding in Na_2O and SO_2 .
- In terms of structure and bonding, explain the difference in the melting point of Na_2O and SO_2 .
- Sketch a graph to show the variation in the melting point of the oxides in the third period of the Periodic Table.
- Describe and explain what you would see when a sample of $\text{Na}_2\text{O(s)}$ and $\text{Al}_2\text{O}_3\text{(s)}$ are separately added to a solution of Universal indicator. Write equations for any reactions that occur.

[8]

(b) Al_2O_3 is used in the dehydration of alcohols to alkenes. Two isomeric alcohols, **A** and **B**, are shown below.

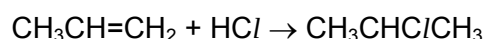


- Draw the structural formula of **one** other alcohol isomeric with **A** and **B**.
- Draw the structural formula of an alkene that is obtained by dehydrating **A**.
- Suggest how alcohols, **A** and **B**, could be distinguished from each other. Give the reagents, conditions and observations with each alcohol.

[4]

(c) Both alcohols and alkenes can be converted to halogenoalkanes.

- State the type of reaction undergone and name a reagent used to convert $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ to $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$.
- Use values of bond energies from the *Data Booklet* to calculate the enthalpy change for the reaction between propene and hydrogen chloride.



[5]

(d) Halogenoalkanes such as $\text{CH}_3\text{CHClCH}_3$ undergo hydrolysis to form alcohols. The relative rates of hydrolysis can be studied by dissolving the halogenoalkane in ethanol and adding aqueous AgNO_3 .

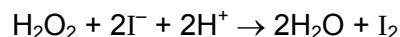
With the aid of the *Data Booklet*, suggest an explanation for the following observations.

Compound	Observations after heating with $\text{AgNO}_3\text{(aq)}$
$\text{CH}_3\text{CHClCH}_3$	ppt forms after 10 minutes
$\text{CH}_3\text{CHBrCH}_3$	ppt forms after 2 minutes
$\text{CH}_3\text{CHICH}_3$	ppt forms immediately

[3]

[Total: 20]

- 7 (a) A study on the kinetics of the reaction between hydrogen peroxide and iodide ions in acidic solution was carried out.



In one experiment, both $[\text{H}_2\text{O}_2]$ and $[\text{H}^+]$ were kept constant at 0.05 mol dm^{-3} and the initial $[\text{I}^-]$ was $0.001 \text{ mol dm}^{-3}$. It was found that the order with respect to $[\text{I}^-]$ is one.

- (i) Sketch the shape of the graph of $[\text{I}^-]$ against time. Use your graph to show clearly how you would determine the order with respect to $[\text{I}^-]$.

The following data was obtained after the experiment was repeated with different $[\text{H}_2\text{O}_2]$ and $[\text{H}^+]$.

Expt	$[\text{H}_2\text{O}_2] / \text{mol dm}^{-3}$	$[\text{H}^+] / \text{mol dm}^{-3}$	relative rate
1	0.05	0.05	1.0
2	0.07	0.05	1.4
3	0.09	0.07	1.8

- (ii) Use the data to deduce the orders with respect to $[\text{H}_2\text{O}_2]$ and $[\text{H}^+]$, explaining your reasoning.
- (iii) Write an expression for the rate equation and state the units of the rate constant. [6]

- (b) The decomposition of hydrogen peroxide is a first order reaction.



The uncatalysed reaction has an activation energy of 79 kJ mol^{-1} . Both platinum and the enzyme catalase act as catalysts, and speed up the reaction.

- (i) Use the data provided to construct a reaction pathway diagram for the uncatalysed reaction.
- (ii) On your sketch in part (b)(i), draw the energy profile of the catalysed reaction and explain how the presence of a catalyst increases the rate of the reaction.
- (iii) What effect will the presence of the catalyst have on the rate constant for this reaction? Explain your answer. [6]

- (c) Three organic compounds, **C**, **D** and **E** each have the same empirical formula CH_2O . The numbers of carbon atoms in their molecules are shown in the table.

compound	number of C atoms
C	1
D	2
E	3

C gives a silver mirror when treated with Tollens' reagent. In **D** and in **E**, the carbon atoms are bonded directly to one another. **D** and **E** each give a brisk effervescence with $\text{Na}_2\text{CO}_3(\text{aq})$. When **E** is heated under reflux with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, the product, **F**, gives an orange–yellow precipitate with 2,4–dinitrophenylhydrazine.

Identify and suggest structures for **C**, **D**, **E** and **F** and explain the reactions involved.

[8]
[Total: 20]

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